A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Adaptation, terminology, succession & population control - Set 1

Mark scheme

AQA spec 3.1.3; The living environment; textbook pp. 72-85

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** The range of an environmental factor within which a species / population can survive (survival highest at the optimum, falling towards the limits) **1** X survives from about pH 5.5 to 8.5 (wide range) – Y only from about pH 6.0 to 8.0 (narrow range) **1** Both have the same optimum (pH 7.0) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **range of tolerance** → the range of conditions in which a species can survive	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** X: 60% survive at pH 6.0 whereas only 5% of Y survive **1** X has a larger gene pool / wider range of tolerance so some individuals are adapted to the new conditions **1** Survivors breed and their offspring are adapted so the population rebuilds – Y is likely to die out locally **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population • **range of tolerance** → the range of conditions in which a species can survive	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reveals the habitat requirements / which abiotic factor is near the edge of tolerance **1** Management can make small beneficial changes to that factor (eg liming, shading, water levels) to keep conditions within the range and increase survival **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **range of tolerance** → the range of conditions in which a species can survive • **abiotic** → non-living / physical factors	2
01.4	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Food supply **1** Pollinators **1** Seed dispersal species **1** Disease / pathogens **1** Predators / competitors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-73; AQA spec 3.1.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2 Species:** a group of closely related organisms that naturally interbreed to produce fertile offspring and are reproductively isolated from other groups (Passer domesticus, house sparrow) – the only 'real' taxon **2 Niche:** the role a species plays in its habitat - how it uses resources and relates to other species (tawny owl: nocturnal carnivore nesting in tree holes eating small mammals) – two species cannot share exactly the same niche in one habitat **2 Community:** all the populations of all species (plants, animals, fungi, bacteria, protists, archaea) living in an area **2 Biome:** a large geographical region with a specific climate and characteristic community, including all areas where it occurs (all tropical rainforests, tundra, savannah) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **Niche** → the role a species plays and the conditions it needs	8
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** An ecosystem is the biotic community plus abiotic factors of one area, relatively self-contained (a particular coral reef) **1** A biome includes all separate ecosystems with the same climate and community type (all coral reefs) – each reef is a different ecosystem within the same biome **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biotic** → living factors / effects of other organisms • **abiotic** → non-living / physical factors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 82-85; AQA spec 3.1.3.2.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Hydrosere **1** Single-celled algae (and floating plants) - arrive on birds / insects / from soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pioneer species** → the first species to colonise bare ground	2

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 5 points below. **1** Pioneers and later plants grow and die – dead organic matter plus sediment washed in fill the pond so water becomes shallower **1** As depth falls rooted plants (pondweeds then emergent reeds with leaves above water) can anchor and colonise – emergents shade submerged plants which die out **1** Sediment / soil accumulates and becomes able to support larger plants – water-tolerant trees (willow, alder) colonise the waterlogged soil **1** Transpiration by trees and further sediment dry the soil and make it aerobic so oak can colonise – the denser canopy shades out smaller plants **1** Each community changes abiotic conditions (depth, light, nutrients, oxygen), making it suitable for the next colonisers which out-compete them – conditions become less extreme and biotic factors increasingly important **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogged** → soil pores filled with water so there is no air • **Transpiration** → water lost as vapour from the leaves • **abiotic** → non-living / physical factors	5
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Starting conditions (bare rock vs open water) differ but each stage makes conditions less extreme and builds soil / organic matter **1** The final community is controlled by the regional climate (temperature, water availability), which is the same for both - climatic climax (eg oak woodland in the UK) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **climax community** → the stable final community **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	2
03.4	1 mark for the point below. **1** Lichens and algae colonise bare rock (extreme temperatures, no soil, little water) – rock fragments and dead organic matter accumulate – mosses, then grasses and ferns as thin soil forms – wind-dispersed trees (birch) then animal-dispersed (oak, beech) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 73-76; AQA spec 3.1.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 3 points below. **1 Primary:** succession on a newly created habitat with no life / no soil (bare rock, new pond, sand) **1 Secondary:** succession restarting after an existing community is disturbed (tree felling, ploughing, fire, end of grazing) **1** Faster because soil already exists and many seeds / roots / organisms are still present so there are fewer delays for colonisation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	3
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Human activity (grazing, mowing, burning, coppicing) regularly removes the later successional species so the climatic climax cannot form **1** The stable community that develops under this repeated disturbance is a plagioclimax **1 Examples:** chalk grassland / hay meadows maintained by grazing or mowing – lowland heath by burning – coppiced woodland – Machair grassland – Broads grazing marshes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deflected succession** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** If the traditional activity stops, secondary succession returns the habitat to scrub / woodland and the plagioclimax species are lost **1** Rainforest and coral reefs are climax communities so conservation aims to minimise human impact **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **plagioclimax** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	2

Do not write outside the box

Question	Marking guidance	Marks
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Extreme environments (deserts, polar) are dominated by abiotic factors – few species tolerate them – populations fluctuate and food webs are simple so changes in one food species have large effects **1** Stable environments allow many species to evolve inter-species relationships – populations are regulated by biotic factors – complex food webs give stability (rainforests, reefs) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **abiotic** → non-living / physical factors • **biotic** → living factors / effects of other organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 76-77; AQA spec 3.1.3.3.	
05.1	**1 mark each** 1 mark for each of the 4 points below. **1** Sigmoidal (S-shaped) growth curve **1** Slow start (few breeding individuals), then rapid / exponential growth (years 4-10) as abundant food and space mean birth rate far exceeds death rate **1** Growth slows as density-dependent factors (intra-species competition for food, disease) raise the death rate **1** Population stabilises / fluctuates around about 920-950 - the carrying capacity - where birth rate equals death rate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support **Do not credit** • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	4
05.2	**1 mark each** 1 mark for each of the 2 points below. **1 Density-dependent:** effect on survival increases with population density - food shortage / intra-species competition, disease spread, predation **1 Density-independent:** chance of survival unaffected by density - drought, flood, volcanic eruption, severe winter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2

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Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Without wolves (natural predators removed) the population is regulated only by starvation / disease at a high density **1** At this density deer over-graze, preventing tree regeneration and damaging habitats / other species – culling keeps numbers below the damaging level (as in Scotland) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Rising prey numbers give predators more food so predator numbers rise – high predation then reduces the prey population **1** Food shortage reduces predators, allowing prey to recover - cyclic fluctuations with predators lagging behind prey **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 78-81; AQA spec 3.1.3.4.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Anchovy r-selected – blue whale k-selected **1** r-selected: mature quickly, produce many young, disperse widely, short lives – adapted to respond rapidly to high death rates / unstable conditions (mice, locusts, greenfly, rabbits) **1** k-selected: mature late, few young with parental care, long-lived – populations stable near carrying capacity (whales, elephants, rhinos) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **carrying capacity** → the largest population the area can support **Do not credit** • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	3
06.2	**1 mark each** 1 mark for each of the 3 points below. **1** Low reproductive rate means losses from hunting / habitat change cannot be replaced quickly **1** An increase in death rate above the low birth rate causes a population crash **1** Recovery takes decades (eg blue whale still far below pre-whaling numbers after the 1986 moratorium) – r-selected species rebound within a few years once exploitation stops **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
06.3	1 mark for the point below. **1** New population = starting population + births + immigrants - deaths - emigrants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
06.4	**1 mark each** 1 mark for each of the 3 points below. **1** The greatest rate of exploitation (harvest) possible without causing long-term population decline **1 Equals the surplus production:** highest when the population is well below carrying capacity where growth rate is fastest (middle of the sigmoid curve) **1** At carrying capacity net growth is zero so any harvest reduces the population – harvesting above MSY causes decline **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **maximum sustainable yield** → the largest catch that can be taken without reducing the stock • **sustainable** → meeting present needs without stopping future generations meeting theirs • **carrying capacity** → the largest population the area can support **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 79-80; AQA spec 3.1.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Succession:** pioneers, seres (hydrosere, lithosere, psammosere), modification of abiotic conditions, climax communities controlled by climate – deflected succession and plagioclimaxes • **Application:** deciding whether to manage (plagioclimax habitats need grazing, mowing, burning, coppicing, pollarding continued) or leave alone (climax habitats) – habitat creation planning the stages – controlling scrub / secondary succession – understanding that new habitats change over time • **Population dynamics:** birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles, r- and k-selection, MSY • **Application:** setting sustainable harvest levels (fisheries, whaling, forestry) – recognising k-selected species need strict protection and recover slowly – deciding when culling is needed (deer without wolves, introduced species) or when releases / captive breeding are needed (low breeding rate) – predicting effects of reintroducing predators • Monitoring population size, distribution, survival rate and age structure to assess success – adaptation and range of tolerance guiding abiotic management • **Evaluation:** uncertainties (natural fluctuations, unknown carrying capacity) – examples (Yellowstone wolves, Scottish deer, chalk grassland) – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **succession** → the gradual change in the community over time • **abiotic** → non-living / physical factors • **deflected succession** → a community held back from its climax by human activity **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (8 marks)	
	Succession is the sequence by which pioneers colonise bare ground, modify the abiotic conditions and are replaced by later seres until a climax community controlled by climate is reached; where grazing, mowing, burning or coppicing halts it, a plagioclimax forms. This tells a conservation manager whether to intervene: chalk grassland, heathland and coppiced woodland are plagioclimaxes that disappear without continued management, whereas a climax habitat is best left alone, and habitat creation must plan the stages, control scrub and manage secondary succession.	
	Population dynamics - birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles and r- and k-selection - inform decisions of a different kind. Maximum sustainable yield sets harvest levels in fisheries, whaling and forestry; k-selected species that breed late and slowly need strict protection and recover slowly; culling may be justified where predators are absent, as with Scottish red deer without wolves, or where introduced species threaten natives, while the reintroduction of wolves to Yellowstone shows how restoring a predator changes vegetation through the food web. Monitoring population size, distribution, survival and age structure measures success, and knowledge of adaptation and ranges of tolerance guides how abiotic conditions are managed.	
	The limits are uncertainty: natural fluctuations, unknown carrying capacities and the difficulty of predicting how communities respond. Ecological understanding cannot remove that uncertainty, but without it conservation decisions - to manage or not, to cull or protect, how much to harvest - are guesses.	
	Weaker answer - Level 2 (5 marks)	
	Succession is when a habitat changes over time from pioneer species to a climax community. Conservationists need to understand this because some habitats like heathland will turn into woodland if they are not managed by grazing or burning. Population dynamics is about how populations grow and are limited by factors like food and predators. This helps conservation because you can work out how many animals can be taken without the population declining, and when a population needs protecting. Monitoring populations shows if conservation is working.	
	What makes the difference	
	• Level 2 explains both concepts correctly and links them to decisions, but generally; Level 3 uses the technical vocabulary (seres, plagioclimax, k-selection, MSY) with named cases (Yellowstone wolves, Scottish deer, chalk grassland).	
	• Level 3 covers more applications (habitat creation, culling introduced species, adaptation and tolerance) and evaluates the uncertainties.	
	• Structure: Level 3 gives each concept its own paragraph with the decisions it informs, then a conclusion.	

Do not write outside the box

Question	Marking guidance	Marks
07.2	1 mark for the point below. **1** Any taxonomic group of organisms classified together by similarity (eg species, genus, family) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-85; AQA spec 3.1.3.	

Do not write outside the box